

How to Monitor and Manage Linux Processes

A program loaded into the memory of a Linux computer becomes a process. Processes need to be managed and monitored because they consume system resources like CPU time, memory and disk space. There are also security and safety implications. Monitoring and managing processes is, therefore, an important function of systems administrators.



A process is a running instance of a launched, executable program. It consists of:

- An address space of allocated memory
- Security properties, including ownership credentials and privileges
- One or more execution threads of program code
- The process state

The environment of a process includes:

- Local and global variables
- A current scheduling context
- Allocated system resources, such as file descriptors and network ports

An existing process duplicates its own address space (fork) to create a new (child) process structure. Every new process has a unique process ID (PID) for tracking and for security. The PID and parent process (PPID) are elements of a new process environment. Any process may create its own child process. All processes are descendants of the first

system process, which is *systemd* in the case of the Red Hat Enterprise Linux 7 system.

Through the *fork* routine, a child process inherits security identities, the previous and current file descriptors, port and resource privileges, environment variables and program code. A child process may then execute its own program code. Normally, a parent process sleeps while the child process runs, setting a request (wait) to be signalled when the child process completes. On exiting, the child process closes or discards its resources and environment; the remainder is referred to as a zombie. The parent, which is signalled awake when the child exits, cleans the remaining structure and continues with executing its own program code.

Process states

In a multi-tasking operating system, each CPU (or CPU core) can work on one process at a single point in time. As a process runs, its immediate requirements for CPU time and